

11 large / 1 T magnetic field applied along body (*allow 'across'*) (1)

r.f. pulse applied (1)
causes hydrogen nuclei / protons (1)
to resonate (1)
(nuclei) return to equilibrium state / after relaxation time (1)
r.f. (pulse) emitted (1)
pulses detected, processed and displayed (1)
resonant frequency depends on magnetic field strength (1)
calibrated non-uniform field enables nuclei to be located (1)

any six points, one mark each B6 [6]

[Total: 6]

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12 (a) a signal can be regenerated